Should we build a sodium-ion battery cell plant in Morocco to supply European carmakers?

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Table of Contents

- Situation
- Complication
- Key Question
- Answer
- Supporting Arguments
- So What
- Requester-provided, not sourceable
- Insufficiently sourced
- Options
- Risks
- Stakeholders
- Adversarial Critique
- Appendix A — Findings
- Appendix B — Stakeholders
- Appendix C — Risk Register
- Sources
- Sourcing Report

Situation

Defer Moroccan plant capital for 12 months; license sodium-ion technology and start OEM sampling first. Gotion started commissioning an LFP cell plant in Kenitra the week of 20 October 2026 ([source](https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/)). Reports put its first phase at 10 or 20 GWh a year. CATL unveiled its Naxtra sodium-ion cell with Changan on 5 February 2026 ([source](https://www.catl.com/en/news/6720.html)). CATL claims up to 175 Wh/kg. One 2026 analysis expects the sodium-ion cost crossover with LFP in 2028–2029 ([source](https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026)). Other sources conflict on current cost. Searches found no European OEM sodium-ion offtake agreements. The Commission's Battery Booster call of 28 July 2026 covers only projects in the European Economic Area ([source](https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en)). COM(2026) 100 of 4 March 2026 promotes Made-in-EU requirements for battery cells ([source](https://single-market-economy.ec.europa.eu/document/download/9bc8eb85-4d43-4025-be7b-c86b9f3648ec_en)). France and Germany reportedly favour a restrictive Made-in-Europe definition ([source](https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/)). Battery Regulation due diligence applies from 18 August 2027 ([source](https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now)). Renewables supplied about 37–40% of Morocco's 2023 electricity ([source](https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test)).

Sources:

- Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/
- CATL and CHANGAN Launch World's First Mass-Production Sodium-Ion Passenger Vehicle — https://www.catl.com/en/news/6720.html
- LFP vs sodium-ion batteries 2026: where the cost crossover actually happens — Earth Energy Log — https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026
- Commission seeks projects to fund under Battery Booster Facility - Internal Market, Industry, Entrepreneurship and SMEs — https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en
- EN EN EUROPEAN COMMISSION Brussels, 4.3.2026 COM(2026) 100 final — https://single-market-economy.ec.europa.eu/document/download/9bc8eb85-4d43-4025-be7b-c86b9f3648ec_en
- European Lawmakers Push to Include Morocco in 'Made in Europe' Industrial Benefits — https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/
- EU Sustainable Batteries Regulation: Where are we now? — https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now
- As Morocco Attracts a $5.6B Battery Plant, Its Coal-Fired Grid Faces a Power Test - Ecofin Agency — https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test

Complication

Gotion began commissioning an LFP cell gigafactory in Kenitra in [October 2026](https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/). A Chinese competitor now occupies the Moroccan site a sodium-ion entrant would target. The Industrial Accelerator Act proposal [requires battery cells to be produced in the EU](https://www.skadden.com/insights/publications/2026/04/european-commission-proposes-industrial-accelerator-act). The act is [not law yet](https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe), and the Council remains split. Battery Booster loans are [open only to projects located in the EEA](https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en). A Moroccan plant is therefore outside the €1.5bn interest-free loan facility. No European OEM offtake agreement or sodium-ion model announcement has been found. Sodium-ion still carries a cost premium to LFP in one 2026 analysis. That analysis [expects the cost crossover in 2028–2029](https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026), not earlier. Sources conflict, and none is primary. CATL already has a [sodium-ion car in mass production](https://www.catl.com/en/news/6720.html) with Changan. Committing factory capital before these facts settle risks a stranded asset. The tension is sharp: waiting cedes ground, and building early risks stranded capital.

Sources:

- Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/
- European Commission Proposes Industrial Accelerator Act: Prioritizing Strategic Sectors, ‘Made in EU’ Requirements and FDI Controls — https://www.skadden.com/insights/publications/2026/04/european-commission-proposes-industrial-accelerator-act
- Made in Europe: the Industrial Accelerator Act and your BOM · GETMILK — https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe
- Commission seeks projects to fund under Battery Booster Facility - Internal Market, Industry, Entrepreneurship and SMEs — https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en
- LFP vs sodium-ion batteries 2026: where the cost crossover actually happens — Earth Energy Log — https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026
- CATL and CHANGAN Launch World's First Mass-Production Sodium-Ion Passenger Vehicle — https://www.catl.com/en/news/6720.html

Key Question

Should the requester fund a Moroccan sodium-ion cell plant before EU origin rules are final?

Answer

The requester should hold Moroccan factory capital until EU origin rules are final. Within 12 months, it should license a cell design, obtain toll-made samples and sign named buyers. That avoids Gotion-scale exposure, reported at $1.3bn for phase one.

No European OEM sodium-ion demand is documented. EEA-only Battery Booster loans and split Council positions leave Moroccan origin status open. Choose Morocco or Iberia only after a sourced landed-cost benchmark exists.

Sources:

- Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/
- Commission seeks projects to fund under Battery Booster Facility - Internal Market, Industry, Entrepreneurship and SMEs — https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en
- Made in Europe: the Industrial Accelerator Act and your BOM · GETMILK — https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe

Supporting Arguments

European demand and cost evidence do not yet justify factory capital before 2030. Research found no European OEM offtake agreements, letters of intent or sodium-ion model announcements. Analysts now expect sodium-ion's cost crossover with LFP in 2028–2029 ([Earth Energy Log](https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026)). Sodium-ion has reached Chinese roads but not yet Europe ([Eleport](https://eleport.com/sodium-ion-vs-lithium-ion/)). A licence and toll-made samples can start OEM qualification now.

Regulatory rules disadvantage Moroccan cells until the Industrial Accelerator Act is final. The proposal states that battery cells must be produced in the EU ([Skadden](https://www.skadden.com/insights/publications/2026/04/european-commission-proposes-industrial-accelerator-act)). The Council remains split on which countries count ([GETMILK](https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe)). Lawmakers are pushing back against a more restrictive line favoured by France and Germany ([Morocco World News](https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/)). Battery Booster loans are open only to projects located in the EEA ([European Commission](https://single-market-economy.ec.europa.eu/news/commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en)). Morocco's grid was only 37–40% renewable in 2023 ([Ecofin](https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test)). EU due diligence duties apply from 18 August 2027 ([CMS](https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now)). Defer factory capital until the final text settles origin.

Chinese incumbents already lead in Kenitra and in sodium-ion technology. Gotion began commissioning its Kenitra gigafactory in October 2026 ([WINS Solutions](https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/)). CATL says Naxtra cells reach 175 Wh/kg in a mass-production passenger car ([CATL](https://www.catl.com/en/news/6720.html)). CATL has also signed European sodium-ion storage deployment deals. Energy Live News reports 3 GWh of sodium-ion storage will be installed in Europe by 2031 ([Energy Live News](https://www.energylivenews.com/2026/11/03/sodium-ion-batteries-are-taking-on-lithium-in-europe/)). A licence with yield guarantees counters this better than building first.

Sources:

- LFP vs sodium-ion batteries 2026: where the cost crossover actually happens — Earth Energy Log — https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026
- Sodium Ion Vs Lithium Ion EV Batteries Differences Explained — https://eleport.com/sodium-ion-vs-lithium-ion/
- European Commission Proposes Industrial Accelerator Act: Prioritizing Strategic Sectors, ‘Made in EU’ Requirements and FDI Controls — https://www.skadden.com/insights/publications/2026/04/european-commission-proposes-industrial-accelerator-act
- Made in Europe: the Industrial Accelerator Act and your BOM · GETMILK — https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe
- European Lawmakers Push to Include Morocco in 'Made in Europe' Industrial Benefits — https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/
- Commission seeks projects to fund under Battery Booster Facility - Internal Market, Industry, Entrepreneurship and SMEs — https://single-market-economy.ec.europa.eu/news/

commission-seeks-projects-fund-under-battery-booster-facility-2026-07-28_en
- As Morocco Attracts a $5.6B Battery Plant, Its Coal-Fired Grid Faces a Power Test - Ecofin Agency — https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test
- EU Sustainable Batteries Regulation: Where are we now? — https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now
- Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/
- CATL and CHANGAN Launch World's First Mass-Production Sodium-Ion Passenger Vehicle — https://www.catl.com/en/news/6720.html
- Sodium-ion batteries are taking on lithium in Europe - Energy Live News — https://www.energylivenews.com/2026/11/03/sodium-ion-batteries-are-taking-on-lithium-in-europe/

So What

Chief Commercial Officer: sign a sodium-ion licence or JV, with toll-made B-samples, within 12 months. Without it, OEM qualification never starts. Volume supply slips to 2030 or later, after Gotion's [10–20 GWh Kenitra phase](https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/) and CATL lock in the first European contracts. Commit no factory capital until the Industrial Accelerator Act's final text treats free-trade-partner cells as Union origin. Then compare Morocco and Iberia on sourced landed-cost and cost-of-capital data. Sign named storage and OEM buyers in parallel.

Sources:

- Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/

Requester-provided, not sourceable

The facts below concern only the requester. No public source can establish them, so this briefing counts none of them as sourced; the requester supplies them, and every statement that rests on them is as reliable as that information.

- Who the requester is (incumbent cell maker, OEM, materials company, investor, startup) and its existing battery manufacturing experience.
- Whether the requester owns or has licensed sodium-ion cell technology, and its current performance and maturity level.
- Available capital budget, target plant capacity (GWh) and required return thresholds or payback horizon.
- Any existing customer relationships, letters of intent or offtake discussions with European carmakers.
- Existing partnerships, JV candidates or prior engagement with the Moroccan government or investment agencies.
- Strategic rationale and board/shareholder constraints (e.g., risk appetite, geographic priorities, timing pressure).

Insufficiently sourced

This format's sourcing policy is strict. The 39 item(s) below did not meet it. They stayed in the analysis that produced this briefing; treat every statement that rests on them as unverified.

- **research item 1**: https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/ is not a trusted domain (host 'www.winssolutions.org' matches no reputation tier).
- **research item 2**: https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026 is not a trusted domain (host 'earthenergylog.com' matches no reputation tier).
- **research item 3**: https://www.catl.com/en/news/6720.html is not a trusted domain (host 'www.catl.com' matches no reputation tier).
- **Research finding 4** — “No evidence was found of contractable European OEM demand for sodium-ion cells (volumes, segments, price points) within a 2027–2030 ramp window.”: no source found (searched for: European carmaker offtake agreements or announced sodium-ion models for Europe. The queries run covered CATL Naxtra, sodium-ion vs LFP cost and Morocco gigafactories, and none returned such evidence.).
- **research item 5**: https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now is not a trusted domain (host 'cms.law' matches no reputation tier).
- **research item 6**: https://www.africanleadershipmagazine.co.uk/morocco-battery-factory-boosts-africas-role-in-global-clean-energy/ is not a trusted domain (host 'www.africanleadershipmagazine.co.uk' matches no reputation tier).
- **research item 7**: https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test is not a trusted domain (host 'www.ecofinagency.com' matches no reputation tier).
- **research item 8**: https://battery-tech.net/battery-markets-news/morocco-to-host-africas-first-battery-gigafactory-in-kenitra/ is not a trusted domain (host 'battery-tech.net' matches no reputation tier).
- **stakeholder item 2**: https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/ is not a trusted domain (host 'www.moroccoworldnews.com' matches no reputation tier).
- **stakeholder item 3**: https://www.moroccoworldnews.com/2026/10/342653/european-lawmakers-push-to-include-morocco-in-made-in-europe-industrial-benefits/ is not a trusted domain (host 'www.moroccoworldnews.com' matches no reputation tier).
- **stakeholder item 4**: https://www.automotiveworld.com/topics/e-mobility/gotion-secures-e100m-to-build-moroccan-battery-plant/ is not a trusted domain (host 'www.automotiveworld.com' matches no reputation tier).
- **stakeholder item 5**: https://www.electrive.com/2026/09/29/industrial-accelerator-act-industry-may-fail-to-meet-battery-demand-in-eu/ is not a trusted domain (host 'www.electrive.com' matches no reputation tier).
- **stakeholder item 6**: https://www.marketscreener.com/quote/stock/STELLANTIS-N-V-117814143/news/Stellantis-to-invest-in-French-sodium-ion-battery-maker-for-EV-output-45728250/ is not a trusted domain (host 'www.marketscreener.com' matches no reputation tier).
- **Stakeholder Other European OEMs (e.g., Renault, VW Group, others) as prospective offtakers**: position not sourced (searched for: Public statements, offtake agreements or sodium-ion model announcements from Renault, VW Group and other European carmakers. Searches covered European sodium-ion plans and Morocco gigafactories, and returned nothing for these OEMs.).
- **stakeholder item 8**: https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-

morocco/ is not a trusted domain (host 'www.winssolutions.org' matches no reputation tier).
- **stakeholder item 9**: https://www.catl.com/en/news/6720.html is not a trusted domain (host 'www.catl.com' matches no reputation tier).
- **stakeholder item 10**: https://dialogue.earth/en/energy/sodium-batteries-are-coming-to-europe/ is not a trusted domain (host 'dialogue.earth' matches no reputation tier).
- **Stakeholder Moroccan power and water suppliers (ONEE, renewable developers, free-zone utilities)**: position not sourced (searched for: Statements from ONEE, Morocco's Energy Transition ministry or renewable developers on supplying power or water to battery plants, beyond a media report on grid strain. That report documents grid conditions, not a stakeholder position.).
- **risk item 1**: https://getmilk.tech/blog/how-much-of-your-product-is-made-in-europe is not a trusted domain (host 'getmilk.tech' matches no reputation tier).
- **risk item 2**: https://www.skadden.com/insights/publications/2026/04/european-commission-proposes-industrial-accelerator-act is not a trusted domain (host 'www.skadden.com' matches no reputation tier).
- **risk item 3**: https://eleport.com/sodium-ion-vs-lithium-ion/ is not a trusted domain (host 'eleport.com' matches no reputation tier).
- **risk item 4**: https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026 is not a trusted domain (host 'earthenergylog.com' matches no reputation tier).
- **risk item 5**: https://carboncredits.com/lithium-prices-surge-amid-strong-demand-forecasts-could-reach-up-to-28000-ton-by-2026-nili/ is not a trusted domain (host 'carboncredits.com' matches no reputation tier).
- **risk item 6**: https://zengineer.blog/blog/deeptech/sodium-ion-battery-vs-lithium-2026-en/ is not a trusted domain (host 'zengineer.blog' matches no reputation tier).
- **risk item 7**: https://www.energylivenews.com/2026/11/03/sodium-ion-batteries-are-taking-on-lithium-in-europe/ is not a trusted domain (host 'www.energylivenews.com' matches no reputation tier).
- **risk item 8**: https://www.forbes.com/sites/michaeltaylor/2025/03/12/europe-loses-ev-battery-race-as-northvolt-files-for-bankruptcy/ is not a trusted domain (host 'www.forbes.com' matches no reputation tier).
- **risk item 10**: https://rdworldonline.com/doe-battman-battery-manufacturing-scale-up is not a trusted domain (host 'rdworldonline.com' matches no reputation tier).
- **risk item 11**: https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test is not a trusted domain (host 'www.ecofinagency.com' matches no reputation tier).
- **risk item 12**: https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now is not a trusted domain (host 'cms.law' matches no reputation tier).
- **risk item 13**: https://www.africanleadershipmagazine.co.uk/morocco-battery-factory-boosts-africas-role-in-global-clean-energy/ is not a trusted domain (host 'www.africanleadershipmagazine.co.uk' matches no reputation tier).
- **Risk RISK-007** (impact evidence): not sourced (searched for: Evidence on skilled-labour shortages or wage inflation for battery manufacturing in Morocco, and their cost impact. No result covered this.).
- **Risk RISK-008** (impact evidence): not sourced (searched for: A benchmark of the cost-of-capital or project-IRR penalty for a battery cell plant outside the EEA that cannot access EU public financing. No result quantified it.).
- **risk item 17**: https://www.digitimes.com/news/a20260213PD231/sodium-ion-automakers-lithium-mercedes-benz-volvo.html is not a trusted domain (host 'www.digitimes.com' matches no reputation tier).
- **risk item 18**: https://eleport.com/new-battery-technologies/ is not a trusted domain (host

'eleport.com' matches no reputation tier).
- **risk item 19**: https://www.marketscreener.com/quote/stock/STELLANTIS-N-V-117814143/news/Stellantis-to-invest-in-French-sodium-ion-battery-maker-for-EV-output-45728250/ is not a trusted domain (host 'www.marketscreener.com' matches no reputation tier).
- **risk item 20**: https://dialogue.earth/en/energy/sodium-batteries-are-coming-to-europe/ is not a trusted domain (host 'dialogue.earth' matches no reputation tier).
- **risk item 21**: https://www.automotiveworld.com/topics/e-mobility/gotion-secures-e100m-to-build-moroccan-battery-plant/ is not a trusted domain (host 'www.automotiveworld.com' matches no reputation tier).
- **Risk RISK-011** (impact evidence): not sourced (searched for: Evidence on the durability of Moroccan battery-sector incentives, and on any precedent where incentive terms changed after investment. No result covered this.).
- **Option OPT-B**: https://www.investing.com/news/economy-news/northvolt-ceo-steps-down-after-company-files-for-bankruptcy-3736651 is not a trusted domain (host 'www.investing.com' matches no reputation tier).

Options

ID	Title	Recommendation	Time horizon
OPT-A	Gated Moroccan pilot line with licensed cell technology and an anchor OEM	recommended	short-term
OPT-B	Full-scale greenfield sodium-ion gigafactory in Morocco now	not-recommended	medium-term
OPT-C	Relocate the sodium-ion plant to an EEA site (Spain or Portugal)	acceptable	medium-term
OPT-D	Defer plant capex; secure technology, offtake and origin rulings first	acceptable	short-term

Risks

ID	Category	Likelihood	Impact	Timeframe
RISK-001	regulatory	medium	major	short-term
RISK-002	strategic	high	severe	short-term
RISK-003	financial	medium	major	medium-term
RISK-004	strategic	high	major	short-term
RISK-005	operational	high	severe	medium-term
RISK-006	regulatory	high	major	short-term
RISK-007	human-capital	medium	moderate	medium-term
RISK-008	financial	very-high	moderate	immediate
RISK-009	technological	medium	major	short-term
RISK-010	strategic	high	major	short-term
RISK-011	geopolitical	low	moderate	medium-term

Stakeholders

Name	Position	Power	Priority
European Commission (DG GROW / DG CLIMA, Battery Booster and Carmaker Accelerator Act)	resistant	high	critical
Governments of France and Germany (Council positions on Made-in-Europe origin)	resistant	high	critical
Spanish Socialist Workers' Party and other pro-Morocco-Morocco-and-its-investment-agency lawmakers	supportive	low	important
Government of Morocco and its investment agency (AMDIE), free-zone authorities, Kenitra/ Tanger-Tech and Casablanca-Settat regional authorities	supportive	high	critical
ACEA (European Automobile Manufacturers Association) and its member OEMs (e.g., Renault, VW Group, Stellantis, Volvo/ Geely, others) as potential offtakers	supportive	medium	important
Stellantis (OEM with Gigafactory in Kenitra, Morocco; 74 GWh; high-tech) and Stellantis Ventures	neutral	high	critical
Other European OEMs and their regional procurement teams (e.g., Renault, VW Group, Volvo/ Geely, others) as potential offtakers	unknown	high	critical
CATL and other Chinese battery cell suppliers (e.g., EVE Energy, Gotion, others) as potential market entrants	resistant	high	important
Thai and other Chinese sodium-ion battery developers (e.g., HiNa Battery, Yadea, others) as potential market entrants	resistant	high	important
Thai and other Chinese sodium-ion battery developers (e.g., HiNa Battery, Yadea, others) as potential market entrants	unknown	medium	monitor
Moroccan power and water suppliers (ONEE, renewable developers, free-zone utilities)	unknown	medium	important

Adversarial Critique

Robustness: low
Critiques: 8 (critical=1, material=6, minor=1)
Revision needed: yes

Steelmanned alternative:
Adopt a capital-light version of OPT-D. Over the next 12 months, sign a sodium-ion licence or JV, have B-samples made on the licensor's line or by a toll manufacturer so OEM qualification starts now, and lock in named storage and OEM buyers. Commit no factory capital until the Industrial Accelerator Act's origin treatment of free-trade-partner cells is fixed in the final text. Then choose between Morocco and Iberia on a sourced landed-cost and cost-of-capital benchmark. This keeps nearly all of OPT-A's qualification benefit without sinking a pilot of likely several hundred million euros outside the EEA, at a time when the leading Western licensor has just collapsed and CATL is already taking European sodium-ion storage demand.

CRIT-001 — temporal-blind-spot (critical)

Steelmanned position:
The brief treats Tiamat as a company building Europe's first sodium-ion gigafactory in Amiens and names it as the first possible licensor for OPT-A. OPT-A requires a licensor that already makes the design at volume and can give ramp-support and yield guarantees. But Tiamat shelved the Amiens project in September 2026 after failing to raise the money. It still has its cells made by subcontractors in China, so it has no volume line of its own that could back yield guarantees. That leaves CATL and HiNa as the only plausible volume licensors. The brief itself classifies them as resistant competitors and says no source shows CATL open to licensing. OPT-A's first dependency may therefore be impossible to meet, and the RISK-010 three-way structure (Tiamat licence, Stellantis offtake, requester builds) is based on an outdated picture.

Implication if true:
If no non-competing licensor with volume production can be signed, OPT-A turns into an unlicensed new-entrant build, which is the Northvolt pattern the brief warns against, or into dependence on a Chinese rival's IP. In either case the recommended option loses its main way of reducing risk.

Suggested revision:
Update the Tiamat stakeholder entry and RISK-010, and make a signed licence term sheet with yield guarantees a precondition for any pilot spend, not something negotiated alongside it.

Counter-evidence:
L'usine devait créer jusqu'à 2000 emplois : incapable de réunir les fonds nécessaires, la start-up Tiamat abandonne son projet de gigafactory de batteries sodium-ion dans la Somme — https://www.usinenouvelle.com/energie/stockage/batteries/lusine-devait-creer-jusqua-2000-emplois-incapable-de-reunir-les-fonds-necessaires-la-start-up-tiamat-abandonne-son-projet-de-gigafactory-de-batteries-sodium-ion-dans-la-somme.5JECNCELKRGRJNAJDEY2C5LX7M.html

CRIT-002 — risk-underestimated (material)

Steelmanned position:
OPT-A and the RISK-002 mitigations rely on stationary-storage buyers for early pilot revenue and on at least 25% non-automotive offtake as a demand hedge. CATL is already locking up European sodium-ion storage demand: a 3 GWh agreement with Intilion in the DACH region, a 5 GWh plan with Alfen, and international deliveries planned from June 2027. A 1-2 GWh new entrant would compete for the same integrators against a scale incumbent. The Industrial

Accelerator Act's storage-origin rules would also apply to publicly supported projects. The storage hedge is therefore weaker than the brief assumes, and the brief rates the residual risk only medium.

Implication if true:
Pilot revenue and the 25% secondary-offtake hedge could fall short. That makes the pilot a pure cost centre and raises the chance that the gates fail with the capital already spent.

Suggested revision:
Add CATL's European sodium-ion storage deals to RISK-004 and RISK-002, and require named storage buyers with pricing before the pilot is approved, not after.

Counter-evidence:
CATL, Intilion target up to 3 GWh of sodium-ion storage in Europe - pv magazine Global — https://www.pv-magazine.com/2026/10/13/catl-intilion-target-up-to-3-gwh-of-sodium-ion-storage-in-europe/

CRIT-003 — temporal-blind-spot (material)

Steelmanned position:
The lithium gate in OPT-A, RISK-003 and So What (a positive IRR at CNY 100,000/t or lower) is presented as a stress test, but it is roughly today's spot price, not a downside case. Chinese carbonate traded around CNY 72,000/t at the end of 2025, rose above CNY 150,000 in January 2026 and is now around CNY 104,000. A real downside test should use late-2025 levels. The 2026 swings also point to something the brief misses entirely: OEMs may value sodium-ion as a hedge against lithium volatility even without an average cost advantage. The brief does not model that rationale either for or against the plant.

Implication if true:
The cost gate could be passed on a near-spot assumption and still leave the plant uneconomic if lithium goes back to 2025 lows. Separately, the volatility-hedge value could be a stronger selling point to OEMs than cost.

Suggested revision:
Reset the IRR gate to CNY 70,000-75,000/t, and add a scenario that values sodium-ion as a hedge against lithium volatility in OEM interviews.

Counter-evidence:
Lithium Prices Analysis Retreat as Chinese Carbonate Falls From Its 2026 Peak — https://bravenewcoin.com/insights/lithium-prices-analysis-retreat-as-chinese-carbonate-falls-from-its-2026-peak

CRIT-004 — overconfidence (material)

Steelmanned position:
The Answer says OPT-A caps exposure at pilot scale and compares it with Gotion's $1.3bn first phase. But the brief itself notes that no capex figure exists for the pilot, and 1-2 GWh is more than many sodium-ion players' first commercial phase. Tiamat's Amiens project was estimated at about €500m for 5 GWh, and it collapsed because it could not close a first phase of nearly €200m, with half the money coming from public grants. A 1-2 GWh line built by a newcomer outside the EEA could plausibly need several hundred million euros. Calling that exposure 'capped' or a 'small share' is not supported.

Implication if true:
The board would be approving a nine-figure commitment that it believes is small. A true pilot below 0.5 GWh, or capacity bought from a toll manufacturer, could provide samples at a fraction of the cost.

Suggested revision:
Give a sourced capex range for 0.3, 1 and 2 GWh pilot options in the Answer and So What, and size the pilot to sample needs rather than to a fixed 1-2 GWh.

Counter-evidence:
Batterie Sodium-Ion à la française : pourquoi Tiamat a renoncé à son usine de batteries près d'Amiens — https://www.generation-nt.com/actualites/tiamat-batterie-sodium-ion-gigafactory-report-2081353

CRIT-005 — risk-underestimated (material)

Steelmanned position:
The RISK-001 mitigation and the OPT-A gate ask for a binding customs-origin opinion under the EU-Morocco Association Agreement. That opinion covers preferential origin, meaning tariff treatment. The Industrial Accelerator Act's Union-origin test, by contrast, is set under the Union Customs Code's non-preferential rules, under which cells made in Morocco are Moroccan, not Union, origin. The gate could therefore be passed while the cells still fail the vehicle-level rule that cells be of Union origin. Inclusion would need a change to the law, which France and Germany oppose. On the current text, exclusion is the base case, not a medium-likelihood risk.

Implication if true:
The origin gate as written gives false comfort. EV channels with public support could be closed to Moroccan cells just as the plant ramps up, because the cell-origin phase begins about six months after the Act enters into force.

Suggested revision:
Change the gate to require a final Industrial Accelerator Act text, or a Council general approach, that explicitly treats free-trade partners' cells as Union origin, and raise RISK-001 likelihood to high.

Counter-evidence:
EN EN EUROPEAN COMMISSION Brussels, XXX […](2026) XXX draft Proposal for a — https://table.media/assets/entwurf-iaa-proposal.pdf

CRIT-006 — dismissed-option (material)

Steelmanned position:
OPT-D was set aside mainly because OEMs will not qualify cells from a supplier without a line. But OPT-A itself depends on a licensed, already-industrialised design. B-samples can come from the licensor's existing line or a toll manufacturer, which is exactly how Tiamat has supplied cells, through Zenergy in China, without any plant of its own. That would start OEM qualification without committing capital to Morocco before the origin rules and cost crossover are settled. The brief's argument that deferral 'cedes 12-18 months' also applies to OPT-A, because its own gates push volume supply to 2029-2030 anyway.

Implication if true:
OPT-D, or a capital-light hybrid with toll-made samples, would capture most of OPT-A's qualification benefit while avoiding the sunk pilot capex and the origin exposure.

Suggested revision:
Add a hybrid OPT-D option (licence plus toll-manufactured B-samples plus origin tracking) and compare it directly with OPT-A on cost and qualification timing.

Counter-evidence:
Tiamat Energy : batteries sodium-ion et gigafactory d'Amiens — https://www.mobilityinthecity.com/f/tiamat-energy/

CRIT-007 — internal-contradiction (material)

Steelmanned position:
Part of the case against OPT-C is that it 'gives up Moroccan cost and incentive advantages'. Yet the same options artefact and its unresolved uncertainties say no capex, unit-cost or payback benchmark between Morocco and Iberia was found. The brief is ranking options on an advantage it says it cannot quantify. Meanwhile, OPT-C is the only option that removes RISK-001, RISK-006 and RISK-008 at once. And ACEA's own study says EU rules of origin remain ambiguous about whether near-shoring partners like Morocco count as equivalent.

Implication if true:
If the Moroccan cost advantage is small or is outweighed by origin and financing penalties, OPT-C, or an Iberian site for the pilot itself, should rank at least equal to OPT-A.

Suggested revision:
Commission a Morocco-versus-Iberia landed-cost and cost-of-capital comparison before choosing the pilot site, and remove the unquantified 'cost advantage' wording from the rationale until it is done.

Counter-evidence:
Industrial Accelerator Act: industry may fail to meet battery demand in EU - electrive.com — https://www.electrive.com/2026/09/29/industrial-accelerator-act-industry-may-fail-to-meet-battery-demand-in-eu/

CRIT-008 — temporal-blind-spot (minor)

Steelmanned position:
The Stellantis entry rests on an undated Reuters item about its Tiamat stake. More recent evidence shows Stellantis anchoring its European cell supply with CATL through a 50 GWh LFP plant in Spain, and Tiamat's own French plant is now shelved. CATL, which already mass-produces sodium-ion Naxtra cells, is therefore the more likely route for any Stellantis sodium-ion volumes. That further weakens the idea that a Moroccan newcomer could become Stellantis's anchor sodium-ion supplier.

Implication if true:
This confirms that RISK-010 is already high likelihood and calls for a non-Stellantis anchor strategy, but it does not change the ranking on its own.

Suggested revision:
Update the Stellantis stakeholder entry with the CATL Spain partnership and the Tiamat suspension, and widen RISK-010's mitigation to give equal priority to non-Stellantis anchors.

Counter-evidence:
CATL's Multi-Country Expansion Redefines Europe's Battery Supply Chain and Workforce • Carbon Credits — https://carboncredits.com/catls-multi-country-expansion-redefines-europes-battery-supply-chain-and-workforce/

Appendix A — Findings

Finding 1

Claim:
Gotion Power Morocco, a subsidiary of Chinese cell maker Gotion High-Tech, began commissioning an LFP cell gigafactory in Kenitra in October 2026. A Chinese competitor with a 20 GWh first phase is therefore already on the Moroccan site that a new sodium-ion entrant would target.

Evidence:
The company said commissioning would start the week of 20 October 2026. The first phase is described as $1.3bn and 20 GWh a year of cells, packs and storage systems, plus cathode materials. Other reports give 10 GWh for phase one and total project values of $5.6–6.8bn, so the capacity and capex figures conflict. The company says output will mainly serve Morocco's car plants and southern Europe.

Source:
Gotion Starts Up Morocco's First Battery Gigafactory — https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/

Finding 2

Claim:
Sodium-ion cells were still reported at a cost premium to LFP in early 2026, with a cost crossover expected around 2028–2029 rather than earlier. Sources disagree on current cost, so the cost-advantage assumption is unproven.

Evidence:
One 2026 analysis puts LFP cells at $84/kWh and sodium-ion at $95–105/kWh, a 13–25% premium, and says the crossover has slipped from 2027 to 2028–2029. Another page claims CATL sodium cells reached $55–70/kWh in Q1 2026, a 35–40% discount to LFP. Neither source is a tier-1 or tier-2 host or a primary data source, so these figures need verification with BNEF, IEA or OEM quotes.

Source:
LFP vs sodium-ion batteries 2026: where the cost crossover actually happens — Earth Energy Log — https://earthenergylog.com/articles/lfp-vs-sodium-ion-2026

Finding 3

Claim:
CATL says its Naxtra sodium-ion cell reaches up to 175 Wh/kg and is in a mass-production passenger car with Changan, unveiled on 5 February 2026. This shows automotive-grade sodium-ion exists, but the evidence comes from a Chinese incumbent that could also compete in Europe.

Evidence:
CATL's press release says it will supply Naxtra cells across Changan's brands and quotes a range above 400 km. The 175 Wh/kg figure and the 'world's first' framing are CATL's own claims, not independently verified.

Source:
CATL and CHANGAN Launch World's First Mass-Production Sodium-Ion Passenger Vehicle — https://www.catl.com/en/news/6720.html

Finding 4

Claim:
No evidence was found of contractable European OEM demand for sodium-ion cells (volumes, segments, price points) within a 2027–2030 ramp window.

Evidence:
The searches run returned only Chinese-market sodium-ion deployments and general cost commentary. They returned no European OEM offtake agreements, letters of intent or sodium-ion model announcements. This is the largest evidence gap for the investment case.

Source:
SOURCE_MISSING — searched for: European carmaker offtake agreements or announced sodium-ion models for Europe. The queries run covered CATL Naxtra, sodium-ion vs LFP cost and Morocco gigafactories, and none returned such evidence.

Finding 5

Claim:
EU Battery Regulation supply-chain due diligence obligations were postponed to 18 August 2027. Carbon-footprint declarations and battery-passport requirements phase in around the same period, so a Moroccan cell plant must be able to document its footprint and supply chain to sell to EU OEMs.

Evidence:
A law-firm summary of Regulation (EU) 2023/1542 says due diligence now applies from 18 August 2027 and covers cobalt, natural graphite, lithium and nickel, with a net-turnover threshold of €40 million. It also says carbon-footprint declarations for EV batteries follow the Commission's methodology acts. Sodium-ion cells use no cobalt or nickel, but the lithium and graphite scope and the footprint rules still affect the plant's data requirements. The source is a secondary law-firm note, not the Official Journal.

Source:
EU Sustainable Batteries Regulation: Where are we now? — https://cms.law/en/gbr/legal-updates/eu-sustainable-batteries-regulation-where-are-we-now

Finding 6

Claim:
Several Chinese battery-supply-chain firms are already investing in Morocco, so a new entrant would compete for talent, utilities and customers with an established Chinese cluster.

Evidence:
Gotion is reported to join BTR, CNGR, Hailiang and Shinzoom in Morocco's battery ecosystem. Other reports describe a CNGR–Al Mada joint venture (COBCO) producing precursor cathode materials and BTR plants in Tangier. This cluster could supply materials locally, but it also competes for the same OEM customers.

Source:
Morocco Battery Factory Boosts Africa's Role In Global Clean Energy - African Leadership Magazine — https://www.africanleadershipmagazine.co.uk/morocco-battery-factory-boosts-africas-role-in-global-clean-energy/

Finding 7

Claim:
Morocco's grid was only partly low-carbon as of 2023, which matters for EU carbon-footprint rules and OEM carbon targets.

Evidence:
Ecofin, citing IRENA and Morocco's Ministry of Energy Transition, reports renewables at about

37–40% of 2023 electricity generation. It describes the grid as coal-dependent and facing a power test from the Gotion plant. A cell plant would likely need dedicated renewable PPAs to reach competitive footprint levels.

Source:
As Morocco Attracts a $5.6B Battery Plant, Its Coal-Fired Grid Faces a Power Test - Ecofin Agency — https://www.ecofinagency.com/news-industry/1609-48731-as-morocco-attracts-a-5-6b-battery-plant-its-coal-fired-grid-faces-a-power-test

Finding 8

Claim:
Morocco has an established automotive export base oriented to the EU, which supports its logistics and customer-proximity case for European OEMs.

Evidence:
Morocco's auto industry posted record 2024 exports of 157 billion dirhams, with roughly 85% of output going to the EU. A separate report notes Renault and Stellantis already operate in Morocco. This is a trade-flow indicator, not proof that battery cells qualify for tariff-free EU access under rules of origin.

Source:
Morocco to Host Africa's First Battery Gigafactory in Kenitra - Battery-Tech Network — https://battery-tech.net/battery-markets-news/morocco-to-host-africas-first-battery-gigafactory-in-kenitra/

Appendix B — Stakeholders

Name	Category	Position	Power	Priority
European Commission (DG GROW / DG ENER, Battery Booster and Clean Mobility Agency (AMDIE), free-zone utilities)	decision-maker	resistant	high	critical
Governments of France, Spain, Germany, Council of the EU, Ministry of Industry and Energy (Morocco), Tangier Free Zone authorities)	decision-maker	resistant	high	critical
EPHA, Battery Passport Consortium, Transport & Environment, Moroccan Network for Sustainable Mobility (RNMD), Tangier Clean Air Coalition)	influencer	supportive	low	important
MEPs of the Committee on Industry, Research and Energy (ITRE) and Committee on International Trade (INTA), Members of the European Parliament from Hauts-de-France and Tangier-Tétouan-Al Hoceïma regions)	gatekeeper	supportive	high	critical
Stellantis (OEM with Mangualde plant), Renault Group (OEM with Tangier plant), Volkswagen Group (OEM with Palmela plant), Ford Otosan (OEM with Craiova plant), Toyota Caetano Portugal (OEM with Ovar plant), Hyundai Motor Europe (OEM with Nošovice plant), Mercedes-Benz (OEM with Jawha plant), BMW Group (OEM with Kenitra plant), Volvo Cars (OEM with Ghent plant), Geely (OEM with Belgium plant), Leapmotor (OEM with Zaragoza plant), Tesla (OEM with Berlin-Brandenburg plant), Dacia (OEM with Tangier

Appendix C — Risk Register

ID	Category	Likelihood	Impact	Timeframe
RISK-001	regulatory	medium	major	short-term
RISK-002	strategic	high	severe	short-term
RISK-003	financial	medium	major	medium-term
RISK-004	strategic	high	major	short-term
RISK-005	operational	high	severe	medium-term
RISK-006	regulatory	high	major	short-term
RISK-007	human-capital	medium	moderate	medium-term
RISK-008	financial	very-high	moderate	immediate
RISK-009	technological	medium	major	short-term
RISK-010	strategic	high	major	short-term
RISK-011	geopolitical	low	moderate	medium-term

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- [research] untrusted: https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/ — host 'www.winssolutions.org' matches no reputation tier
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- [stakeholder] 'Other European OEMs (e.g., Renault, VW Group, others) as prospective offtakers' SOURCE_MISSING — Public statements, offtake agreements or sodium-ion model announcements from Renault, VW Group and other European carmakers. Searches covered European sodium-ion plans and Morocco gigafactories, and returned nothing for these OEMs.
- [stakeholder] untrusted: https://www.winssolutions.org/gotion-kenitra-battery-gigafactory-morocco/ — host 'www.winssolutions.org' matches no reputation tier

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- [stakeholder] 'Moroccan power and water suppliers (ONEE, renewable developers, free-zone utilities)' SOURCE_MISSING — Statements from ONEE, Morocco's Energy Transition ministry or renewable developers on supplying power or water to battery plants, beyond a media report on grid strain. That report documents grid conditions, not a stakeholder position.
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- [risk] RISK-007 .impact_evidence SOURCE_MISSING — Evidence on skilled-labour shortages or wage inflation for battery manufacturing in Morocco, and their cost impact. No result covered this.
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- [risk] RISK-011 .impact_evidence SOURCE_MISSING — Evidence on the durability of Moroccan battery-sector incentives, and on any precedent where incentive terms changed after investment. No result covered this.

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- [adversarial] CRIT-001 cites an untrusted domain: https://www.usinenouvelle.com/energie/stockage/batteries/lusine-devait-creer-jusqua-2000-emplois-incapable-de-reunir-les-fonds-necessaires-la-start-up-tiamat-abandonne-son-projet-de-gigafactory-de-batteries-sodium-ion-dans-la-somme.5JECNCELKRGRJNAJDEY2C5LX7M.html — host 'www.usinenouvelle.com' matches no reputation tier
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